Solvers

Grid and spectral

- JAX-CFD
- PhiFlow
- XLB
- PICT
- Warp-NS
- Exponax
- INS.jl
- OpenFOAM

Finite element

- FEniCS
- Firedrake
- JAX-FEM
- torch-fem
- TopOpt.jl
- deal.II

Interface

Tesseract

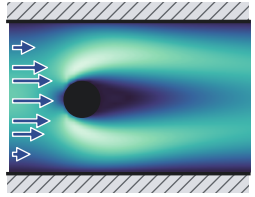
`apply(x) → y`

`vjp(x, v) → g`

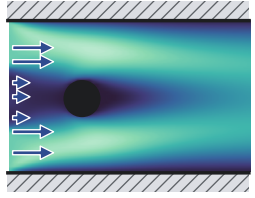
one container per solver

Tasks

Forward solve



Optimized inflow



plus H, S and F3 tasks

Results

2D Navier–Stokes, checks per suite

	Fwd	Cost	Grad	Opt.
JAX-CFD	—	✓	✓	—
PhiFlow	✓	✓	✓	—
INS.jl	✓	✓	✓	✓
XLB	✓	✓	✓	✓
PICT	—	—	✓	✓
Warp-NS	—	✓	—	—
Exponax	—	✓	✓	✓
OpenFOAM	—	✓		

✓ pass — partly not applicable